Retail Sales Analytics — README

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Source file: README.md

Retail Sales Data Analytics and AI-Based Sales Prediction

Project overview

This project analyzes retail transactions and trains a machine-learning model to estimate sales amount. It includes data cleaning checks, exploratory analysis, visualizations, and a Random Forest regression model.

Important: The included 600-row dataset is synthetic/sample data generated for learning and demonstration. It is not real customer or company data. Replace it with an approved real dataset if your internship instructions require one.

Files

- `retail_sales_dataset.csv` — sample dataset (600 rows)
- `retail_sales_analysis.py` — end-to-end analysis and model training script
- `project_report.pdf` — project report
- `requirements.txt` — Python dependencies
- PNG files — report charts

How to run

1. Install Python 3.10+.
2. Open a terminal in this folder.
3. Install dependencies: `pip install -r requirements.txt`
4. Run: `python retail_sales_analysis.py`

Methods

- Data quality checks and date feature extraction
- Group-by analysis by product category, region and sales channel
- One-hot encoding for categorical features
- Train/test split (80%/20%)
- Random Forest Regressor
- MAE, RMSE and R^2 evaluation

Results from this generated sample dataset

- Records: 600
- Total sales: ■11,582,841.27
- Average sales per record: ■19,304.74

- Total units sold: 2,686
- Test MAE: ■2,460.64
- Test RMSE: ■6,277.29
- Test R^2: 0.955

Metrics can change if the dataset, Python libraries, or model settings are changed.